section status ✗ not reviewed, nothing here is approved *figure status* ● on file, in a committed result file ◐ partial, unfinished or unscored ○ idea only, nothing run

color, the input family the gene's own sequence (genotype only) the strain's measured phenotype as input (transductive) permuted or nothing fitted (control)

regime shaded band: transductive, needs a measurement of the strain itself; unshaded: inductive, the genotype alone

Morphology from node-level measurements. Can a cell-level phenotype (CalMorph) be predicted from the genotype, and does the strain's own measured gene-level modality (expression, proteome) help when given as input? Panels a and b describe the data; c is the inductive arm; d, e and f need the strain's measurement (transductive); g is more training data for the measured modalities, not a panel of its own.

a ✗ ●

The strain sets
three single-deletion panels and their overlaps: morphology 4,695, expression 1,554, proteome 4,476; morph. and expr. 1,439, morph. and prot. 4,281, all three 1,321. Counts, no statistic.

results/morph_v24_dataset.json

b ✗ ●

The target: 116 moving features
per-feature replicate reliability (122 wild-type profiles against 4,718 mutants) as a histogram; the 0.5 threshold; ceiling = sqrt(reliability), mean 0.847 on the moving set, 0.611 on all 278.

results/morphology_feature_ceiling.csv, calmorph_moving_features.json

c ✗ ●

Genotype only (inductive)
bars: ceiling; sequence ridge and neighbor average; the CGT off the perturbed CLS (twelve split seeds, window mean, sd). Pearson per feature across held-out strains, moving features.

results/morph_v24_readout.json (M_cls, M_pool)

transductive: the strain's own measurement as input

d ✗ ◐

Measured modality as input
paired by split seed: the strain's own proteome (expression) as input against another strain's (permuted, control) and against genotype only on the same strains; twelve seeds; beside the ridge from the same measurement.

morph_v24_readout.json (CM_prot, CM_protperm, MP_cls; CMP_expr, CMP_exprperm); in flight

e ✗ ○

Per feature
held-out Pearson of the best measured-input arm against each feature's reliability; which features a measured modality reaches and which stay at the floor. Dots, 116 features, one seed by rule.

a per-feature dump of the CM_prot checkpoint; not written yet

f ✗ ●

The conditioning triangle
the same contrast on the other two sides: proteome given the strain's expression +0.054 (11 of 11), expression given its proteome +0.033 (10 of 11), both below their ridge; morphology is the third vertex.

results/joint_checkpoint_readout.json (v20), the covariation files

g ✗ ○

More training strains for the measured modalities: the 1,011 natural isolates (Peter 2018)
Not a morphology panel: no CalMorph exists for these strains. The idea is to add them as extra training data for the expression head (Caudal 2024 pan-transcriptome, built, 943 isolates) and the proteome head (Muenzner 2024, 796 isolates, next build), about 1,000 strains whose expression differs far more between strains than the deletion panels do, and then ask whether the deletion-strain heads, and through them the transductive morphology arms in d, improve. Open question first: how an isolate's genotype enters a model that takes a set of deleted genes. **Idea only; likely not in time for this round.**

the expansion document, next-build band; no result file